Solutions to Exercises from *High-Dimensional Probability*

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Appetizer: Using Probability to Cover a Set

Exercise 0.1. (Two variance formulas)

Solution.

1.
$$\mathbb{E}\|Z-\mathbb{E}Z\|_2^2 = \mathbb{E}\sum_{i=1}^n (Z_i-\mathbb{E}Z_i)^2 = \sum_{i=1}^n \mathbb{E}(Z_i-\mathbb{E}Z_i)^2 = \sum_{i=1}^n \mathbb{E}Z_i^2 - (\mathbb{E}Z_i)^2$$
$$= \mathbb{E}\sum_{i=1}^n Z_i^2 - \sum_{i=1}^n (\mathbb{E}Z_i)^2 = \mathbb{E}\|Z\|_2^2 - \|\mathbb{E}Z\|_2^2$$

Other way:

$$\mathbb{E}\|Z-\mathbb{E}Z\|_2^2 = \mathbb{E}(Z-\mathbb{E}Z)^\top(Z-\mathbb{E}Z) = \mathbb{E}Z^\top Z - \mathbb{E}(Z^\top \mathbb{E}Z) - \mathbb{E}((\mathbb{E}Z)^\top Z) + \mathbb{E}(\mathbb{E}Z)^\top(\mathbb{E}Z)$$
$$= \mathbb{E}\|Z\|_2^2 - 2(\mathbb{E}Z)^\top(\mathbb{E}Z) + (\mathbb{E}Z)^\top(\mathbb{E}Z) = \mathbb{E}\|Z\|_2^2 - \|\mathbb{E}Z\|_2^2$$

2.
$$\mathbb{E}\|Z-\mathbb{E}Z\|_2^2 = \sum_{i=1}^n \mathbb{E}(Z_i-\mathbb{E}Z_i)^2 = \sum_{i=1}^n \mathbb{E}Z_i^2 - (\mathbb{E}Z_i)^2 = \frac{1}{2}\sum_{i=1}^n 2\mathbb{E}Z_i^2 - 2(\mathbb{E}Z_i)^2$$
$$= \frac{1}{2}\sum_{i=1}^n \mathbb{E}Z_i^2 + \mathbb{E}(Z_i')^2 - 2(\mathbb{E}Z_i)(\mathbb{E}Z_i')$$
$$= \frac{1}{2}\sum_{i=1}^n \mathbb{E}Z_i^2 + \mathbb{E}(Z_i')^2 - 2(\mathbb{E}Z_iZ_i') \quad \text{from independence}$$
$$= \frac{1}{2}\sum_{i=1}^n \mathbb{E}(Z_i^2 + (Z_i')^2 - 2Z_iZ_i') = \frac{1}{2}\sum_{i=1}^n \mathbb{E}(Z_i-Z_i')^2 = \frac{1}{2}\mathbb{E}\|Z-Z'\|_2^2$$

□

Exercise 0.2. (Expectation minimizes the mean squared error) The question has a typo and it should say:
$$\mathrm{Var}(X) = \min_{a\in\mathbb{R}} \mathbb{E}(X-a)^2$$

Solution. First lets check the scalar version:
$$\min_{a\in\mathbb{R}} \mathbb{E}(X-a)^2 = \min_{a\in\mathbb{R}} \mathbb{E}X^2 - 2a\mathbb{E}X + a^2 = \min_{a\in\mathbb{R}} \mathrm{Var}(X) + ((\mathbb{E}X)-a)^2 = \mathrm{Var}(X)$$

Now lets prove the euclidean vector version:
$$\min_{a\in\mathbb{R}^n} \mathbb{E}\|Z-a\|_2^2 = \min_{a\in\mathbb{R}^n} \mathbb{E}\|Z\|_2^2 - 2\langle \mathbb{E}Z, a\rangle + \|a\|_2^2$$
$$= \min_{a\in\mathbb{R}^n} \mathbb{E}\|Z-\mathbb{E}Z\|_2^2 + \|\mathbb{E}Z\|_2^2 - 2\langle \mathbb{E}Z, a\rangle + \|a\|_2^2 \quad \text{from Exercise 0.1}$$

□

Exercise 0.3. (Variance of a sum)

Solution.

$$\mathbb{E}\|\sum_{j=1}^{k} Z_j\|_2^2 = \mathbb{E}\sum_{i,j=1}^{k} Z_i^\top Z_j = \mathbb{E}(\sum_{j=1}^{k}\|Z_j\|_2^2 + \sum_{i\neq j} Z_i^\top Z_j) = \sum_{j=1}^{k}\mathbb{E}\|Z_j\|_2^2 + \sum_{i\neq j}\mathbb{E}Z_i^\top Z_j$$

where $Z_i^\top Z_j$ is also a linear combination so:

$$\mathbb{E}Z_i^\top Z_j = (\mathbb{E}Z_i)^\top \mathbb{E}Z_j = 0^\top 0 = 0$$

□

Exercise 0.4. (Balancing vectors)

Solution. Similar to the Maurey method. We can assume random variables σ_i such that $p_{\sigma_i}(-1) = p_{\sigma_i}(1) = 0.5$. And now we just prove that the expected value is less than the radius. Using the last exercise:

$$\mathbb{E}\|\sum_{i=1}^{n}\sigma_i x_i\|_2^2 = \sum_{i=1}^{n}\mathbb{E}\|\sigma_i x_i\|_2^2 = \sum_{i=1}^{n}\mathbb{E}\|x_i\|_2^2 \leq n$$

$$\implies \exists\sigma_i \quad \|\sum_{i=1}^{n}\sigma_i x_i\|_2^2 \leq n$$

□

1 A Quick Refresher on Analysis and Probability

Exercise 1.1.

Solution. □

Exercise 1.2.

Solution.

$$f(x) := \max_i f_i(x) \quad f_i \text{ are convex functions}$$

$$f(\lambda x + (1-\lambda)y) = \max_i f_i(\lambda x + (1-\lambda)y) \leq$$

$$\max_i \lambda f_i(x) + (1-\lambda)f_i(y) \leq \max_i \lambda f_i(x) + \max_i (1-\lambda)f_i(y) = \lambda f(x) + (1-\lambda)f(y)$$

□

Exercise 1.3.

Solution.

1. induction on the definition
2. definition of expected value for discrete random variable

□

Exercise 1.4.

Solution. We know $T \subseteq \text{conv}(T)$, therefore $\sup_{x\in T} f(x) \leq \sup_{x\in \text{conv}(T)} f(x)$.
Also we have for some $\epsilon > 0$ and $S := \sup_{x\in \text{conv}(T)} f(x)$ that $\exists y \in \text{conv}(T)\ f(y) > S - \epsilon$. Now $y = \sum_{i=1}^m \lambda_i x_i$ for $x_i \in T, \lambda \geq 0, \sum_{i=1}^m \lambda_i = 1$ which also means $\lambda_i \leq 1$. By convexity of f and Jensen's inequality we have:

$$f(y) \leq \sum_{i=1}^{m} \lambda_i f(x_i) \leq \max_{1\leq i\leq m} f(x_i) \leq \sup_{x\in T} f(x)$$

So

$$[\forall \epsilon > 0 \quad S - \epsilon < f(y) \leq \sup_{x\in T} f(x)] \implies \sup_{x\in \text{conv}(T)} f(x) \leq \sup_{x\in T} f(x)$$

□

Exercise 1.5.

Solution. case $n = 2$: $x = (x_1, x_2)$, $x_i \in [0, 1]$:

$$x = (1 - x_2)(x_1(1, 0) + (1 - x_1)(0, 0)) + x_2(x_1(1, 1) + (1 - x_1)(0, 1))$$

$$= (1 - x_2)(x_1)(1, 0) + (1 - x_2)(1 - x_2)(0, 0) + x_2(x_1)(1, 1) + x_2(1 - x_1)(0, 1)$$

conjecture: For each $v = (v_1, \cdots, v_n) \in \{0, 1\}^n$ define:

$$\lambda_v := \prod_{i=1}^{n} [(v_i)(x_i) + (1 - v_i)(1 - x_i)]$$

proof by induction: We can verify for $n = 1, 2$ as above. Assume this works for $n = k - 1$, meaning any point in $[0, 1]^{k-1}$ can be written as $x = \sum_{v\in\{0,1\}^{k-1}} \lambda_v v$. We can prove now for $n = k$ by taking a chosing one of its dimensions and letting a hyper-surface orthogonal to that point create a $k - 1$ dimensional cube. now we have □

Exercise 1.6.

Exercise 1.7.

Exercise 1.8.

Solution. We want to prove that with probability less than or equal to n there exists an independent group of size more than $2 \log_2 n$. We have $n \geq 7$, therefore $\lfloor 2 \log_2 n \rfloor > 4$. So we have the size $k \geq 5$.

□

Exercise 1.9.

Exercise 1.10.

Exercise 1.11.

Exercise 1.12.

Solution.

1. An inequality that relates to different norms is Holder's inequality so we try to use that. We define $r := \frac{q}{p}$, where its conjugate is $r' := 1/(1 - \frac{p}{q}) = \frac{q}{q-p}$. We take the r-norm of X^p and r'-norm of 1 to see:
$$\|X^p\|_1 \leq \|X^p\|_r \|1\|_{r'} = \|X^p\|_r$$
$$\implies \mathbb{E}|X^p| \leq (\mathbb{E}|X|^q)^{p/q} \implies answer$$

   For $0 \leq p < 1$ we should have used Jensen instead of Holder.

2. Recall $\|X\|_{Lp} = (\mathbb{E}|X|^p)^{1/p}$. For continuous X with distribution f we have $\mathbb{E}|X|^p = \int_D |x|^p f(x)\, dx$, where D is the domain of f (range of X). Take $\alpha = \frac{p+q}{2}$. Now if we had $f(x) = |x|^{-1-\alpha}$ then $|x|^{p-1-\alpha}$ will converge (its exponent is ≤ -1) and $|x|^{q-1-\alpha}$ will diverge. But we need to make sure f is a valid density function, i.e. $\int_D f(x)\ dx = 1$. To make things simpler we take $X \geq 1$ so we can remove the absolution value functions. Now we have $int_1^\infty x^{-1-\alpha} = [x^{-\alpha}/-\alpha]_1^\infty = 1/\alpha$. So we must define $f_X(x) := \alpha x^{-1-\alpha}$.
   Also for a discrete example we could create an infinite sum that converges diverges when its elements are gone to the powers more than p and converges otherwise.

□

Exercise 1.13.

Solution. □

2 Concentration of Sums of Independent Random Variables

Exercise 2.1. (Products of i.i.d. random variables do not concentrate)

$$X_1, \cdots, X_n \in \text{Uniform}([0,1])$$
$$Y_n := X_1 \cdots X_n$$

Solution. Since X_i are independent:

$$\mathbb{E}[Y_n] = \prod_{i=1}^n \mathbb{E}[X_i] = \prod_{i=1}^n \int_0^1 x\ dx = 2^{-n}$$

$$\mathbb{P}[Y_n \geq \mathbb{E}Y_n] = \mathbb{P}[Y_n \geq 2^{-n}] \geq \mathbb{P}(Y_1 \geq \frac{1}{2} \wedge \cdots \wedge Y_n \geq \frac{1}{2}) = (0.5)^n$$

We also have:

$$\log Y_n = \sum_{i=1}^n \log X_i, \quad \log X_i \in (-\infty, 0]$$

$$\mathbb{P}[Y_n \geq \mathbb{E}Y_n] = \mathbb{P}(\sum_{i=1}^n \log X_i \geq -n \log 2) \leq$$

???

□

Exercise 2.2. (Gaussian tails: a lower bound)

Solution.

$$\mathbb{P}(g \geq t) = \int_t^\infty \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx, \quad \frac{d}{dt}\mathbb{P}(g \geq t) = -\frac{1}{\sqrt{2\pi}} e^{-t^2/2}$$

$$\frac{d}{dt}\frac{t}{t^2+1} \cdot \frac{1}{\sqrt{2\pi}} e^{-t^2/2} = \frac{1}{\sqrt{2\pi}}\big(\frac{1-2t^2-t^4}{(t^2+1)^2}\big) e^{-t^2/2}$$

defining $h(t) := \mathbb{P}(g \geq t) - \frac{t}{t^2+1} \cdot \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$ we can see that

$$\frac{d}{dt} h(t) = \frac{-2}{(t^2+1)^2\sqrt{2\pi}} e^{-t^2/2}$$

. We have $h'(t) < 0$ for all t. and $\lim_{t\to\infty} h'(t) = 0$. So the function $h(t)$ is strictly decreasing. We can also verify that $h(0) > 0$ and $\lim_{t\to\infty} h(t) = 0$, so the function cannot become negative at any point t.

□

Exercise 2.3. (Mills ratio)

Exercise 2.4. (Truncated Gaussian moments)

Solution.

1.

$$\mathbb{E}[g\mathbf{1}_{g>t}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^\infty x\mathbf{1}_{x>t} e^{-x^2/2}\, dx = \frac{1}{\sqrt{2\pi}} \int_t^\infty x e^{-x^2/2}\, dx = \frac{1}{\sqrt{2\pi}} [-e^{-x^2/2}]_t^\infty = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$$

2.

$$\mathbb{E}[g^2\mathbf{1}_{g>t}] = \frac{1}{\sqrt{2\pi}} \int_t^\infty x^2 e^{-x^2/2}\, dxvol$$

Integration by parts $u := x$ and $v := e^{-x^2/2}$ gives:

$$\leq \frac{1}{\sqrt{2\pi}}\big(te^{-t^2/2} + \int_t^\infty e^{-x^2/2}\, dx\big) \leq \frac{1}{\sqrt{2\pi}}\big(te^{-t^2/2} + \frac{1}{t}e^{-t^2/2}\big) = \frac{1}{\sqrt{2\pi}}(t + \frac{1}{t})e^{-t^2/2}$$

□

Exercise 2.5. (Completing the proof of Hoeffding inequality)

Solution. Taylor expansion □

Exercise 2.6. (Gaussian tail by the exponential moment method)

Solution. for any $t \geq 0$:

$$\mathbb{P}(g \geq t) = \mathbb{P}(e^{\lambda g} \geq e^{\lambda t}) \leq e^{-\lambda t}\mathbb{E}[e^{\lambda g}]$$

$$= e^{-\lambda t} \int_{-\infty}^\infty e^{\lambda x} \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx = e^{-\lambda t} \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} e^{\frac{\lambda^2 - (x-\lambda)^2}{2}}\, dx$$

$$= e^{-\lambda t + \lambda^2/2} \int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}} e^{\frac{-(x-\lambda)^2}{2}}\, dx = e^{-\lambda t + \lambda^2/2}$$

$$\text{minimize by setting } \lambda = t \implies \mathbb{P}(g \geq t) \leq e^{-t^2/2}$$

□

Exercise 2.7. (Small ball probability)

Solution.

$$\mathbb{P}(\sum_{i=1}^{N} X_i \leq \epsilon N) = \int_{\sum_{i=1}^{N} x_i \leq \epsilon N} f(x_1, \cdots, x_N)\, dx = \int_{\sum_{i=1}^{N} x_i \leq \epsilon N} f(x_1) \cdots f(x_N)\, dx \leq K^N \int_{\sum_{i=1}^{N} x_i \leq \epsilon N} 1\, dx$$

So, we need the volume of a simplex $C := \{\sum_{i=1}^{N} x_i \leq \epsilon N \mid 0 \leq x_i\}$. Using the transform $T(x_1, \cdots, x_N) := \epsilon N \cdot (x_1, \cdots, x_N)$ we have that $\text{Vol}(C) = \epsilon^N N^N \text{Vol}(\{\sum_{i=1}^{N} x_i \leq 1 \mid 0 \leq x_i\})$.

$$\text{Vol}(C) = \epsilon^N N^N$$

attempt for a more typical argument:

$$\mathbb{P}(\sum_{i=1}^{N} X_i \leq \epsilon N) = \mathbb{P}(e^{\lambda \sum_{i=1}^{N} X_i} \leq e^{\lambda \epsilon N}) \leq e^{-\lambda \epsilon N} \prod_{i=1}^{N} \mathbb{E}[e^{\lambda X_i}] = e^{-\lambda \epsilon N} \prod_{i=1}^{N}$$

□

Exercise 2.8. (An MGF comparison inequality)

Solution. We have:

$$\int_{\mathcal{X}} x f_X(x)\, dx = pa + (1-p)b := m$$

, where p is the probability of $Y = a$.

$$\implies p = \frac{b-m}{b-a}$$

We should utilize the convexity of the $e^{\lambda x}$ function. So, a line intersecting it at $x = a$ and $x = b$ lies above the function for all $x \in (a, b)$:

$$e^{\lambda x} \leq \frac{b-x}{b-a} e^{\lambda a} + \frac{x-a}{b-a} e^{\lambda b} \quad \forall x \in [a, b]$$

$$\implies \mathbb{E}e^{\lambda X} \leq \frac{b - \mathbb{E}X}{b-a} e^{\lambda a} + \frac{(\mathbb{E}X) - a}{b-a} e^{\lambda b} = \frac{b-m}{b-a} e^{\lambda a} + \frac{m-a}{b-a} e^{\lambda b} = pe^{\lambda a} + (1-p)e^{\lambda b} = \mathbb{E}e^{\lambda Y}$$

□

Exercise 2.9. (Hoeffding lemma)

Solution.

1. For a r.v. X we have the random variable $Z := X - \mathbb{E}X \in [a', b']$, where $a' = a - \mathbb{E}X$ and $b' = b - \mathbb{E}X$. Now $b' - a' = b - a$ and we only need to prove the lemma for Z while $\mathbb{E}Z = 0$.

   For any $C = b - a > 0$ we can define $Z := \frac{X}{C}$ and so $Z \in [a', b']$ with $b' - a' = 1$. Now if the lemma holds for Z then we have:

   $$\mathbb{E}e^{\lambda(Z - \mathbb{E}Z)} \leq \exp\big(\lambda^2 (b'-a')^2/8\big) \implies \mathbb{E}e^{\frac{\lambda}{C}(X - \mathbb{E}X)} \leq \exp\left(\frac{\lambda^2}{C^2}(b-a)^2/8\right)$$

   , which holds for all $\lambda' := \lambda / C \in \mathbb{R}$.

Now assuming X and Z both have zero mean and $b-a=1$, we know from last exercise:

$$\mathbb{E}e^{\lambda X} \leq \mathbb{E}e^{\lambda Z}$$

and if the result holds for Z then $\mathbb{E}e^{\lambda Z} \leq \exp\big(\lambda^2(b-a)^2/8\big)$. The RHS does not directly depend on the intervals intermediate values.

2.

$$K(\lambda) := \log \mathbb{E}e^{\lambda X}, \quad X \in \{a, b\}$$
$$\implies K(0) = \log 1 = 0$$

also in this case:

$$K'(0) = \frac{\mathbb{E}[Xe^{\lambda X}]}{\mathbb{E}[e^{\lambda X}]}\bigg|_{\lambda=0} = \frac{\mathbb{E}X \cdot 1}{\mathbb{E}1} = \mathbb{E}X$$

and

$$K''(\lambda) = \frac{(\mathbb{E}X^2e^{\lambda X})(\mathbb{E}e^{\lambda X}) - (\mathbb{E}Xe^{\lambda X})^2}{(\mathbb{E}e^{\lambda X})^2}$$
$$= \frac{(pa^2e^{\lambda a} + (1-p)b^2e^{\lambda b})(pe^{\lambda a} + (1-p)e^{\lambda b}) - (pae^{\lambda a} + (1-p)be^{\lambda b})^2}{(pe^{\lambda a} + (1-p)e^{\lambda b})^2}$$
$$= \frac{p(1-p)e^{\lambda(a+b)}(a^2+b^2) - 2p(1-p)abe^{\lambda(a+b)}}{(pe^{\lambda a} + (1-p)e^{\lambda b})^2}$$

From $\mathbb{E}X = 0$ we have $pa + (1-p)b = 0 \implies p = \frac{b}{b-a} = b$. So:

$$K''(\lambda) = \frac{-abe^{\lambda(a+b)}(a^2+b^2) + 2a^2b^2e^{\lambda(a+b)}}{(be^{\lambda a} + -ae^{\lambda b})^2}$$
$$= \frac{-abe^{\lambda(a+b)}\overbrace{((a^2+b^2) - 2ab)}^{=(b-a)^2}}{(be^{\lambda a} + -ae^{\lambda b})^2} = \frac{-abe^{\lambda(a+b)}}{(be^{\lambda a} + -ae^{\lambda b})^2} \leq \frac{1}{4}$$

, where the last inequality is from $(A+B)^2 \leq 2AB$.

Now we see use Taylor's theorem with Lagrange form of the remainder:

$$K(\lambda) = K(0) + \lambda K'(0) + \frac{\lambda^2}{2!}K''(\xi), \quad \forall \xi \in (0, \lambda) \text{ or } (\lambda, 0)$$
$$\implies K(\lambda) \leq \frac{\lambda^2}{8} \implies \mathbb{E}e^X \leq e^{\frac{\lambda^2}{8}}$$

Also from $\mathbb{E}X = 0$ we know $a \leq 0 \leq b$ so $K''(\lambda) \geq 0$ and therefore $K(0)$ is at a local minimum and we have a convexity here.

□

Exercise 2.10. (Hoeffding inequality for bounded random variables)

Solution. We have:

$$\mathbb{P}(\sum_{i=1}^{N}(X_i - \mathbb{E}X_i) \geq t) = \mathbb{P}(e^{\lambda\sum_{i=1}^{N}(X_i-\mathbb{E}X_i)} \geq e^{\lambda t}) \leq e^{-\lambda t}\mathbb{E}e^{\lambda\sum_{i=1}^{N}(X_i-\mathbb{E}X_i)}$$

$$= e^{-\lambda t}\prod_{i=1}^{N}\mathbb{E}e^{\lambda(X_i-\mathbb{E}X_i)} \leq e^{-\lambda t}e^{\frac{\lambda^2\sum_{i=1}^{N}(b_i-a_i)^2}{8}} = e^{\frac{\lambda^2\sum_{i=1}^{N}(b_i-a_i)^2}{8}-\lambda t}$$

now we choose λ to minimize the bound:

$$\frac{d}{d\lambda}\frac{\lambda^2\sum_{i=1}^{N}(b_i-a_i)^2}{8} - \lambda t = \frac{\sum_{i=1}^{N}(b_i-a_i)^2}{4}\lambda - t \implies \lambda^* = \frac{4t}{\sum_{i=1}^{N}(b_i-a_i)^2}$$

$$\implies \frac{\sum_{i=1}^{N}(b_i-a_i)^2}{4}\lambda^* - t = 0 \implies \frac{\sum_{i=1}^{N}(b_i-a_i)^2}{4}(\lambda^*)^2 - t\lambda^* = 0$$

$$= \frac{\sum_{i=1}^{N}(b_i-a_i)^2}{8}(\lambda^*)^2 + \frac{\sum_{i=1}^{N}(b_i-a_i)^2}{8}(\lambda^*)^2 - t\lambda^*$$

$$\implies \frac{\sum_{i=1}^{N}(b_i-a_i)^2}{8}(\lambda^*)^2 - t\lambda^* = -\frac{\sum_{i=1}^{N}(b_i-a_i)^2}{8}(\lambda^*)^2 = -\frac{2t^2}{\sum_{i=1}^{N}(b_i-a_i)^2}$$

□

Exercise 2.11. (Chernoff inequality: left tails)

Solution. For $\lambda \geq 0$ and $0 < t \leq \mu$

$$\mathbb{P}(S_N \leq t) = \mathbb{P}(e^{-\lambda S_N} \geq e^{-\lambda t}) \leq e^{\lambda t}\mathbb{E}e^{-\lambda S_N}$$

$$= e^{\lambda t}\prod_{i=1}^{N}(p_i e^{-\lambda} + (1-p_i)e^0) = e^{\lambda t}\prod_{i=1}^{N}(1 + p_i(e^{-\lambda}-1))$$

$$\leq e^{\lambda t}\prod_{i=1}^{N}\exp\Big(p_i(e^{-\lambda}-1)\Big) = e^{\lambda t}\exp\left((e^{-\lambda}-1)\sum_{i=1}^{N}p_i\right) = e^{\lambda t}\exp\Big((e^{-\lambda}-1)\mu\Big)$$

$$= \exp\Big((e^{-\lambda}-1)\mu + \lambda t\Big)$$

.

Now we minimize $K(\lambda) \coloneqq (e^{-\lambda}-1)\mu + \lambda t$ over $\lambda \geq 0$:

$$K'(\lambda) = -\mu e^{-\lambda} + t \implies \lambda^* = -\ln\frac{t}{\mu} \text{ which is non-negative by assumption } 0 < t \leq \mu$$

$$K(\lambda^*) = (\frac{t}{\mu}-1)\mu - \ln\left(\frac{t}{\mu}\right)^t = t - \mu - \ln\left(\frac{t}{\mu}\right)^t$$

$$\implies \mathbb{P}(S_N \leq t) \leq e^{-\mu}(\frac{e\mu}{t})^t$$

□

Exercise 2.12. (Reverse Chernoff)

Solution.

$$\mathbb{P}(S_N = t) = \binom{N}{t}(\frac{\mu}{N})^t(\frac{N-\mu}{N})^{N-t}$$

□

Exercise 2.13.

Exercise 2.14.

Exercise 2.15.

Exercise 2.16.

Exercise 2.17.

Exercise 2.18.

Exercise 2.19.

Exercise 2.20.

Exercise 2.21. (Boosting randomized algorithms)

Solution. Take the correctness of the algorithm as the Bernoulli r.v. $X \in \{0,1\}$ with $\mathbb{P}(X = 1) = \frac{1}{2} + \epsilon$. The correctness after running N times and taking majority vote is:

$$\mathbb{P}(S_N \leq \frac{N}{2}) \quad \text{while } \mathbb{E}S_N = \frac{N}{2} + N\epsilon$$

$$= \mathbb{P}(S_N - \mathbb{E}S \leq -N\epsilon) \leq \exp\left(-\frac{2N^2\epsilon^2}{N}\right) = \exp\bigl(-2N\epsilon^2\bigr)$$

We want

$$\exp\bigl(-2N\epsilon^2\bigr) \leq \delta \iff N \geq \frac{1}{2\epsilon^2}\ln\frac{1}{\delta}$$

□

Exercise 2.22. (Absolute moments of the normal distribution)

Solution.

1.

$$g \sim N(0,1)$$

$$\mathbb{E}|g|^p = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}}|x|^p e^{\frac{-x^2}{2}}\,dx = \frac{2}{\sqrt{2\pi}}\int_0^{\infty} x^p e^{\frac{-x^2}{2}}\,dx$$

we set $t \coloneqq x^2/2$, and so $\frac{dt}{dx} = x$:

$$= \frac{2}{\sqrt{2\pi}}\int_0^{\infty} x^{p-1}e^{\frac{-x^2}{2}}\,dt = \frac{2}{\sqrt{2\pi}}\int_0^{\infty} 2^{\frac{p-1}{2}}(\frac{x^2}{2})^{\frac{p-1}{2}}e^{-t}\,dt$$

$$= \frac{2^{\frac{p+1}{2}}}{\sqrt{2\pi}}\int_0^{\infty} t^{\frac{p-1}{2}}e^{-t}\,dt$$

$$= \frac{2^{\frac{p+1}{2}}}{\sqrt{2\pi}}\Gamma(\frac{p+1}{2}) = \frac{2^{\frac{p}{2}}}{\sqrt{\pi}}\Gamma(\frac{p+1}{2})$$

2. By Stirling's formula we have:

$$\Gamma(\frac{p-1}{2}+1) \sim \sqrt{2\pi\frac{p-1}{2}}(\frac{p-1}{2e})^{\frac{p-1}{2}}(1+o(1))$$

$$\implies \frac{2^{\frac{p}{2}}}{\sqrt{\pi}}\Gamma(\frac{p+1}{2}) \sim \sqrt{2\frac{(p-1)^p}{e^{p-1}}}(1+o(1)) =$$

$$(\mathbb{E}|g|^p)^{1/p} \sim \sqrt{2\frac{(p-1)}{e^{\frac{p-1}{p}}}}(1+o(1))^{1/p}$$

$$\sim \sqrt{\frac{p}{e}}(1+o(1)) \quad \text{as } p \to \infty$$

□

Exercise 2.23. (Subgaussian MGF requires zero mean)

Solution.

$$\mathbb{E}\exp(\lambda X) \le \exp\big(K^2\lambda^2\big)$$

$$\implies f(\lambda) \le g(\lambda), \quad f(0) = g(0) = 1$$

$$\implies f'(0) \le g'(0) \text{ while } f'(0) = \mathbb{E}X \text{ and } g'(0) = 0$$

$$\implies \mathbb{E}X \le 0$$

doing the same for $-\lambda$ gives $\mathbb{E}X \ge 0$.

□

Exercise 2.24. (Examples of subgaussian distributions)

Solution.

1. (Constant).

$$\mathbb{E}e^{X^2/K^2} \le 2 \implies \int_{\mathcal{C}} e^{c^2/K^2} f_X(x)\,dx \le 2$$

$$\implies e^{c^2/K^2} \le 2 \implies K^2 \ge c^2/\ln 2$$

2. (Bounded). First we clarify: X is bounded a.s. means that $\|X\|_\infty = \sup_x\{\mathbb{P}(X \le x) = 1\}$ exists. Therefore:

$$\frac{X^2}{K^2} \le \frac{\|X\|_\infty^2}{K^2} \text{ a.s.}$$

$$\implies \mathbb{E}e^{\frac{X^2}{K^2}} \le e^{\frac{\|X\|_\infty^2}{K^2}}$$

So for a K that $e^{\frac{\|X\|_\infty^2}{K^2}} \le 2$ we will definitely have $\mathbb{E}e^{\frac{X^2}{K^2}} \le 2$ and so $\|X\|_{\psi_2} \le K$. So we find this bound for such K:

$$e^{\frac{\|X\|_\infty^2}{K^2}} \le 2 \implies K \ge \|X\|_\infty/\sqrt{\ln 2}$$

, by taking $K = \|X\|_\infty/\sqrt{\ln 2}$ and that $\|X\|_{\psi_2} \le K$, we get $\|X\|_{\psi_2} \le \|X\|_\infty/\sqrt{\ln 2}$

3. (Rademacher).

$$\mathbb{E}e^{\frac{X^2}{K^2}} \leq 2 \implies e^{\frac{1}{K^2}} \leq 2$$
$$\implies K \geq \frac{1}{\sqrt{\ln 2}} \implies \|X\|_{\psi_2} = \frac{1}{\sqrt{\ln 2}}$$

4. (Normal) we have $X \sim N(0, \sigma^2)$

$$\mathbb{E}e^{\frac{X^2}{K^2}} \leq 2 \implies \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\frac{(2-K^2)x^2}{2K^2}}\, dx \leq 2$$

Setting $\frac{(2-K^2)x^2}{2K^2} = -\frac{t^2}{2} \implies \frac{(K^2-2)}{K^2}x^2 = t^2 \implies dt = \sqrt{\frac{(K^2-2)}{K^2}}\, dx$, we get:

$$\sqrt{\frac{K^2}{K^2-2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\frac{-t^2}{2}}\, dt \leq 2 \implies \sqrt{\frac{K^2}{K^2-2}} \leq 2$$
$$3K^2 - 8 \geq 0 \implies K \geq \sqrt{8/3} \text{ or } K \leq -\sqrt{8/3}$$

5. (Bernoulli)

$$\mathbb{E}e^{X^2/K^2} = pe^{1/K^2} + 1 - p \leq 2$$
$$\implies e^{1/K^2} \leq \frac{1}{p} + 1 \implies K^2 \geq \frac{1}{\sqrt{\ln\left(\frac{1}{p}+1\right)}}$$

□

Exercise 2.25. (Examples of non-subgaussian distributions) Explain why the exponential, Poisson, geometric, chi-squared, Gamma, Cauchy, and Pareto distributions are not subgaussian.

Solution.

- (Exponential) For $t \geq 0$:

$$X \sim \text{Exp}(\lambda) \implies \mathbb{E}e^{tX} = \int_0^{\infty} e^{tx}\lambda e^{-\lambda x}\, dx$$

, which diverges for $t \geq \lambda$. Also there exists $t \geq 1 \geq 0$:

$$\mathbb{P}(X \geq t) = e^{-\lambda t} \geq e^{-\lambda t^2}$$

Actually it is interesting to see that for t near zero it does behave like (or lighter) than sub-Gaussian.

- (Poisson) For $X \sim \text{Poisson}(\lambda)$, and using Stirling's formula:

$$\mathbb{P}(X \geq t) \geq \mathbb{P}(X = t) = e^{-\lambda}\frac{\lambda^t}{t!} = \frac{e^{-\lambda}}{\sqrt{2\pi t}}(\frac{e\lambda}{t})^t(1 + o(1)) \quad \text{as } t \to \infty$$

, so there is a t such that:

$$\mathbb{P}(X \geq t) \geq C_1 t^{-\frac{1}{2}-t} \geq C_2 t^{-t} \geq C_3 e^{-t\ln t} \geq C_4 e^{-t^2}$$

- (Geometric) For $X \sim \text{Geometric}(\lambda)$, we have:

$$\mathbb{P}(X \geq t) = (1-\lambda)^{t-1} = e^{t\ln(1-\lambda)}$$

- (Chi-squared)

□

Exercise 2.26. (Subgaussian characterization: MGF of X^2)

Solution. $\implies$: take $K = \|X\|_{\psi_2} < \infty$, which exists. Then:

$$\mathbb{E}e^{\lambda^2 X^2} = \mathbb{E}[(e^{X^2/K^2})^{\lambda^2 K^2}]$$

now since $\lambda^2 K^2 \in [0,1]$, we have $(\cdot)^{\lambda^2 K^2}$ is a concave function. So with Jensen's inequality we have:

$$\leq (\mathbb{E}e^{X^2/K^2})^{\lambda^2 K^2} \leq 2^{\lambda^2 K^2} \leq e^{\lambda^2 K^2}$$

$\impliedby$: We take $\lambda = 1/(2K)$ and get the result.

□

Exercise 2.27. (Subgaussian characterization: almost stochastic dominance)

Solution.

1. First we assume X is sub-Gaussian. Then:

$$\mathbb{P}(|X| \geq t) \leq 2e^{-ct^2/\|X\|_{\psi_2}^2} \quad \text{and} \quad \frac{4Kt}{t^2+K^2}\frac{1}{\sqrt{2\pi}}e^{-\frac{t^2}{2K^2}} \leq 2\mathbb{P}(K|g| \geq t) \;\; K > 0$$

   We can show that $\frac{2Kt}{t^2+K^2} \leq 1$. We should solve the following for $K > 0$:

$$2e^{-ct^2/\|X\|_{\psi_2}^2} \leq \frac{4Kt}{t^2+K^2}\frac{1}{\sqrt{2\pi}}e^{-\frac{t^2}{2K^2}}$$

$$\implies \exp\left((\frac{1}{2K^2} - \frac{c}{\|X\|_{\psi_2}^2})t^2\right) \leq \frac{1}{\sqrt{2\pi}}$$

2.

□

Exercise 2.28. (Subgaussian characterization: convex dominance)

Exercise 2.29. (Hoeffding inequality for bounded random variables)

Solution. $X_i \in [a_i, b_i]$ and $\mathbb{E}X_i = 0$. We get that $\|X\|_{\psi_2}$ □